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Build with AI (Google) • Track 3

PHC Pulse

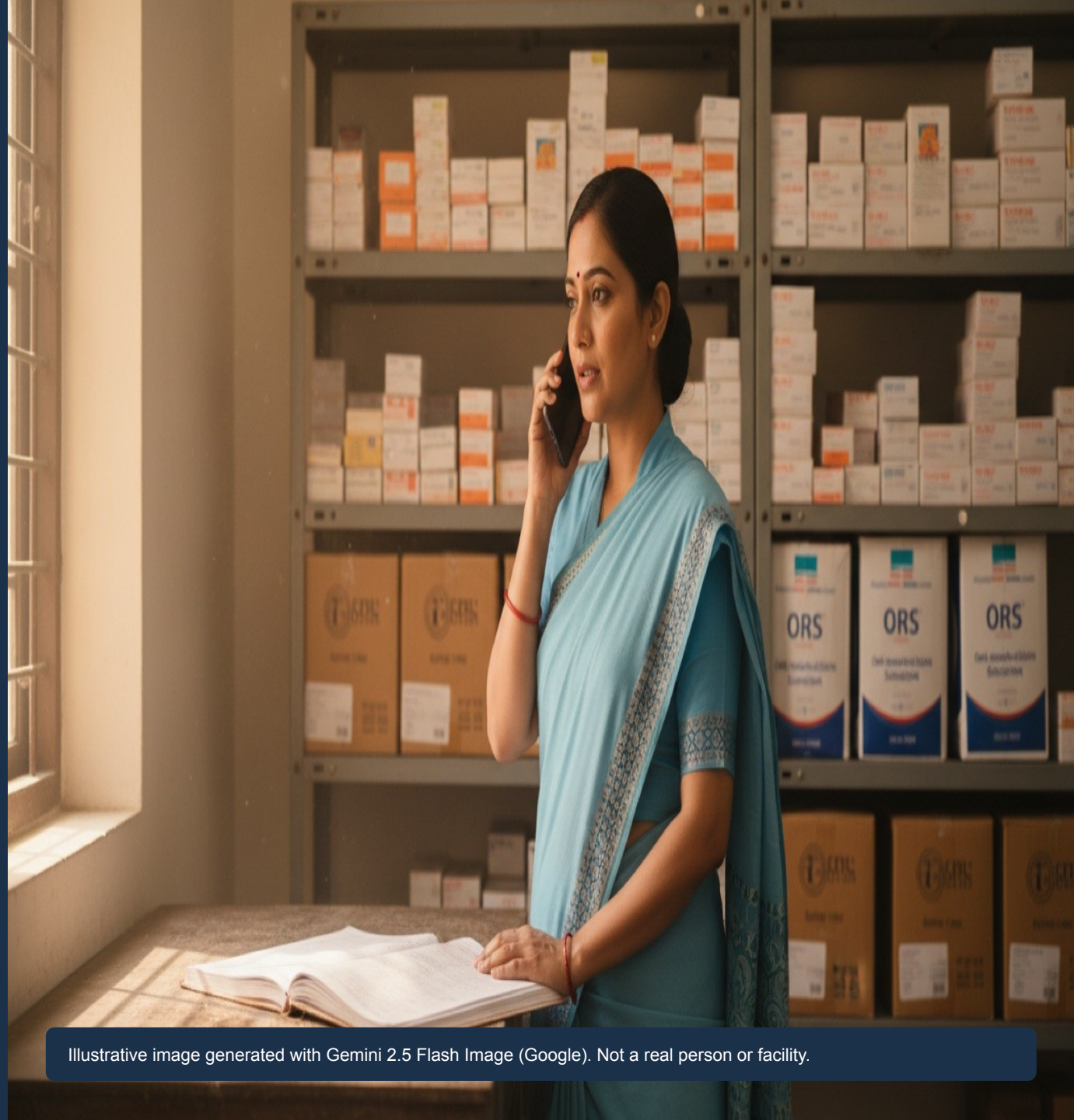
No stock-out here while the same medicine expires next door.

Voice and photo stock reporting, demand forecasts and optimised transfers for Primary Health Centres, approved by the District Health Officer.

AI-ASSISTED | OFFICER-APPROVED | STATE-READY

A concept proposal and working prototype by Spontom Enterprise Private Limited

DPIIT Recognised • Visakhapatnam • September 2026



Illustrative image generated with Gemini 2.5 Flash Image (Google). Not a real person or facility.

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Stock-outs and expiry sit side by side.

31,882 PHCs are the first point of care [2]; their stock data sits in registers, spreadsheets and separate systems [1].

One nurse, one register, one flood week.

3.6×	2.5	450
ORS + zinc use	days of ORS left	sachets expiring nearby

Kavitha copies stock into a paper register each evening. The district sees it days later, after the ORS has run out.

SYNTHETIC SCENARIO

COUNT — Paper register at the PHC

›

REPORT — Periodic and late

›

? — Nobody sees the surplus next door

›

INDENT — Fixed cycle, 10-day lead time

FOCUSED PROBLEM

Districts cannot see in time which PHC will run out, or which nearby PHC can spare the same medicine.

Consequence Patients buy medicines or travel further; surplus expires on another shelf. [1]

[1] Track 3 brief; PHC Pulse PRD §2. [2] MoHFW, Health Dynamics of India 2022–23. Card figures: synthetic scenario A (PHC Melur, Madurai).

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From a spoken report to an approved transfer.

Staff report in their language; AI structures and forecasts; an optimiser proposes; the officer decides.

01 REPORT
Voice in Tamil, Hindi or English, a register photo or text. Gemini extracts; staff confirm.

02 SEE + FORECAST
Days of cover, readiness score, 8-week forecast and possible-surge signals.

03 OPTIMISE + APPROVE
OR-Tools proposes transfers; Gemini explains; the DHO approves each one.

PROPOSED TRANSFER — AWAITING DHO

451 ORS sachets • PHC Thirumangalam II → PHC Melur

1. Melur: 2.5 → 10.3 days of cover (stock-out risk 99.8%)
2. Donor keeps 50.6 days; its safety stock is 45 days
3. Uses 450 sachets that expire on 6 Dec 2026
4. 54 min drive (Google Routes API) • 7.8 stock-out days avoided
5. **Gemini flags: Melur report is 30 h old; surge unverified**

Gemini 2.5 Flash • explain_transfer_v1 — APPROVE

Real optimiser and Gemini output from the local build (30 Sep 2026) on synthetic data. Quantities come from OR-Tools; Gemini cannot change them.

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One journey from report to receipt.

Observed data, then AI interpretation, then a recommendation, then a human decision (PRD §35).

1 **SPEAK** — Voice, photo, text or form

2 **EXTRACT** — Gemini → validated JSON

3 **CONFIRM** — Staff edit, then save

4 **SEE** — Cover + readiness

5 **FORECAST** — 8 weeks + surge check

6 **OPTIMISE** — OR-Tools transfers

7 **APPROVE** — DHO → dispatch → receipt

TIER 0 • AUTOMATIC

Cover, risk, forecasts and surge checks recompute on every confirmed report.

TIER 1 • DHO REVIEW

The officer verifies surge signals and approves or rejects each proposed transfer.

TIER 2 • STATE

Gaps local surplus cannot cover go to a state indent; cross-district moves follow state policy.

DESIGN ASSUMPTION

The 45-day donor safety stock, 30-day target cover and readiness weights are defaults to set with each state.

Readiness = stock cover 40 • beds 20 • staff 20 • reporting freshness 10 • surge stability 10 (PRD §13; weights per state).

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Design for the PHC nurse; serve the district.

Two roles in the product: PHC Staff and District Health Officer (PRD §7). No patient-level data.



KAVITHA • STAFF NURSE (PERSONA)

PHC Melur • 6 beds • 7 staff • one paper register

01	**PHC COLLEAGUES** Pharmacist / MO	Same app and facility scope: report stock, update beds, confirm receipts.
02	**DISTRICT OFFICER** DHO • approver	Needs trusted cover, risk and transfers for 32 facilities, not phone calls.
03	**DRUG WAREHOUSE** District supply	10-day indent lead time; forecast shortfalls show what to send first.
04	**STATE HEALTH DEPT** Configures + scales	Sets weights, adapter and transfer policy; sees aggregates, not raw data.

CONSTRAINT → DESIGN RESPONSE

LITTLE TIME	voice report in seconds
LANGUAGE	Tamil • Hindi • English
PAPER REGISTER	photo, Gemini reads

Illustrative image generated with Gemini 2.5 Flash Image (Google). Persona from PRD §7 and scenario A; facility figures are synthetic.

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Scenario A: a flood week in Madurai, end to end.

Tamil Nadu • Madurai • post-flood diarrhoea • ORS stock-out risk (PRD §43, synthetic data).

PHC MELUR • TAMIL VOICE

Stock report

"ORS 40 packets… Amoxicillin finished" → 3 items matched, confirmed.

›

INSTANTLY

Days of cover

ORS 2.5 days • stock-out risk ≈100% • readiness 61/100.

›

DISTRICT VIEW

Possible surge

ORS + zinc 3.6× normal across 8 facilities; DHO verifies.

›

OPTIMISER

12 transfers proposed

8,142 ORS sachets on Routes drive times; each needs approval.

AFTER APPROVAL

Melur receives 885 sachets from two donors.

Both PHCs are notified in their own language:

+ **DHO approves 451 + 434 sachets**

✓ **Donors stay at ≥ 45 days of cover**

↗ **Receipt closes the loop in the app**

WHY OFFICERS TRUST IT

Every record carries source, time and model version; unverified surges and old reports are flagged.

All figures from the synthetic dataset (local run, 30 Sep 2026). The Tamil sentence is the demo script from docs/SUBMISSION.md.

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Complements state systems; closes the transfer gap.

Existing systems solve real parts of the problem. PHC Pulse plugs in through adapters instead of replacing them.

APPROACH	WHAT IT CONTRIBUTES	WHERE PHC PULSE ADDS
Paper register + periodic report	Familiar, offline, the facility's official record	Photo of the same register becomes structured stock
DVDMS / state drug software	Procurement, warehouse and facility inventory (NHM FDSI) [3]	Voice capture at PHCs; optimised facility-to-facility transfers
HMIS	National service-delivery reporting [4]	Daily stock decisions, not monthly indicators
IDSP / IHIP	Disease surveillance and outbreak reporting [5]	Medicine-use surge as a possible signal for the DHO to verify
PHC Pulse	Voice/photo → forecast → OR-Tools transfer → DHO approval; state adapters	Prototype on synthetic data; not yet field-validated

DISTINCTIVE CAPABILITY

A spoken PHC report becomes an explained, officer-approved transfer.

[3] NHM Free Drugs Service Initiative guidelines. [4] HMIS. [5] IHIP. Features of state deployments vary; PHC Pulse integrates rather than replaces.

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Bounded AI on Google Cloud, not a chatbot.

Every Google integration sits behind an adapter with a labelled fallback, so the journey never silently breaks.

01	**CAPTURE**	Speech-to-Text (ta/hi/en) • register photo • text • form • staff confirmation
02	**AI EXTRACTION**	Gemini 2.5 Flash on Vertex AI • schema-validated JSON • unknown codes rejected
03	**CANONICAL DATA**	TN / UP / OD adapters → canonical schema • BigQuery ledger
04	**DECISION LAYER**	Shared 8-week forecast • surge rules • OR-Tools min-cost flow on Routes times
05	**DELIVERY**	PHC app (EN/HI/TA, TTS) • DHO dashboard with Maps • 3 Cloud Run services

GOOGLE AI INTEGRATION MAP

- Gemini: extraction, register reading, explanations
- Speech, TTS, Translation • OR-Tools on Maps Routes
- BigQuery ledger • BQML ARIMA_PLUS (1,536 series)

NOT YET **Firestore alerts, Firebase Auth, Model Registry**

REAL-VS-DEMO RULE

Live	Probed at start-up; the badge shows live only if the call passes
Fallback	Labelled demo mode (rule parser, haversine); never silent
Freshness	Every record has source + time; reports > 72 h are flagged

Shared forecast, synthetic data only: WAPE 0.134–0.145 vs 0.172–0.189 for same-week-last-year across the three states (model card).

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AI proposes. People confirm and approve.

Reported data, AI interpretation and recommendations are kept separate and labelled (PRD §35).

HUMAN APPROVAL

NOTHING MOVES WITHOUT THE DHO

Staff confirm	AI-extracted stock is editable; nothing reaches the ledger unconfirmed.
DHO approves	Every transfer and every surge notice needs officer review first.

NEVER

- **Store patient-level or personal health data**
- **Invent quantities: unclear values stay blank**
- **Present a surge as a confirmed outbreak**

SAFETY BY DESIGN

SCHEMA-BOUND	Gemini returns schema-validated JSON; unknown drug codes are rejected.
OPTIMISER DECIDES	OR-Tools sets quantities; Gemini explains but cannot change them.
SAFETY STOCK	Donors never drop below their safety cover (45 days by default).
PROVENANCE	Every AI output stores model name, version and prompt version.
STAFF PRIVACY	Check-in, not tracking; dashboards show aggregate presence only.
DATA STAYS IN STATE	States share models and aggregate metrics, never facility data.

Register photos and voice recordings are kept in Cloud Storage for audit; retention limits are a pilot decision (PRD §52).

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Prove fewer stock-out days, then scale by state.

The first proof is not app usage. It is medicine on the shelf when patients arrive.

01
REPORTING
Daily stock by voice or photo

›

02
VISIBILITY
Cover the DHO trusts

›

03
ACTION
Approved transfers before stock-out

›

04
PATIENT SIGNAL
Fewer stock-out days, less expiry

MEASURE WITH

Stock-out days per tracer medicine [3]
Near-expiry units moved vs expired
Warning lead time before stock-out
Reporting rate + extraction accuracy

VALIDATION GATES

1 **DISTRICT PILOT**
1 district, 20–40 PHCs; real opening stock via the state adapter

2 **COMPARE FAIRLY**
Pilot vs matched blocks; baseline stock-out days from state records

3 **CLAIM CAREFULLY**
Synthetic results show feasibility only; publish real accuracy and limits

OWNERSHIP + SCALE

Owner	State Health Dept / State NHM
Channel	DHO + district drug warehouse
Gate	Data access • language QA • DHO capacity

BUDGET ROUTE

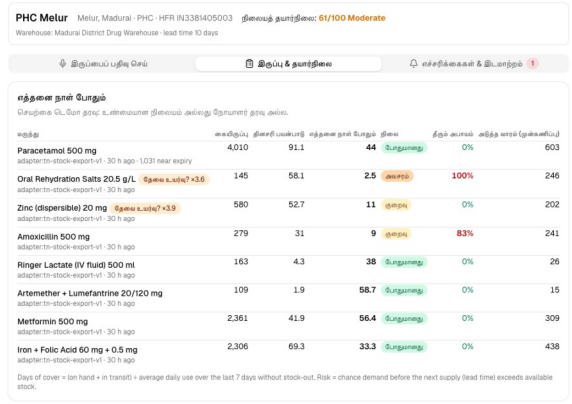
NHM Free Drugs Service Initiative support for IT-enabled drug supply chains, via the State PIP. [3]

Scale path (PRD §50): district pilot → state integration with drug-distribution software → shared model registry across states.

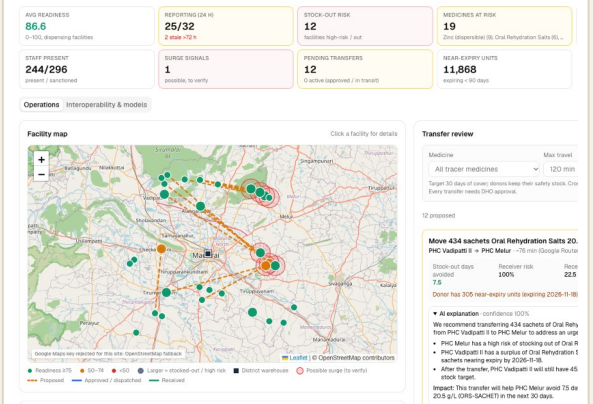
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Working today, end to end.

Three Cloud Run services in asia-south1 (Mumbai). All data is synthetic and labelled as such.



PHC staff app • Tamil • PHC Melur stock



DHO dashboard • map, KPIs, transfer review

TRY IT

PHC staff app	https://health-resource-network-phc-web-847963771142.asia-south1.run.app
DHO dashboard	https://health-resource-network-officer-dashboard-847963771142.asia-south1.run.app
API (OpenAPI)	https://health-resource-network-api-847963771142.asia-south1.run.app/docs
Source	https://github.com/ipavanreddy/health-resource-network

EVALUATION ALIGNMENT

25% — AI / TECHNICAL EXECUTION
Gemini extraction and register reading, OR-Tools on Routes times, validated JSON

20% — PROBLEM-SOLUTION FIT
Report → cover → forecast → warning → transfer → approval in one journey

20% — DEPTH & REACH ACROSS INDIA
3 state adapters, EN/HI/TA voice, one shared model with per-state accuracy

20% — DEPLOYABILITY & SCALABILITY
3 Cloud Run services; onboarding a new state is a config file

15% — IMPACT POTENTIAL
31,882 PHCs [2]; outcome = FDSI stock-out-day indicator [3]

Screenshots: local build, 30 Sep 2026, synthetic data. Locally the map uses the OpenStreetMap fallback.

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Challenge evidence & public health sources

Numbered references correspond to citations in the core proposal.

[1] Track 3 brief and PHC Pulse PRD

Challenge statement and product requirements (docs/PRD.md), incl. demo scenarios §43 and scale plan §50. Synthetic dataset: data/generator/synth.py.

LOCAL EVIDENCE

[2] MoHFW (2024), Health Dynamics of India 2022–23

Formerly Rural Health Statistics. 31,882 PHCs and 47,932 staff nurses at PHCs as of 31 March 2023 (PIB release, 9 Sep 2024).

OPEN SOURCE ↗

[3] NHM Free Drugs Service Initiative guidelines

Supports IT-enabled supply chains such as DVDMS; stock-out days as an indicator; redistribute near-expiry drugs to facilities in need; PIPs.

OPEN SOURCE ↗

[4] Health Management Information System (HMIS)

MoHFW portal for facility-level service-delivery reporting and publications.

OPEN SOURCE ↗

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Platform, registry & technology sources

Systems PHC Pulse complements, and the Google technology it runs on.

[5] **Integrated Health Information Platform (IHIP)**

MoHFW platform for disease surveillance under IDSP; where possible-surge signals would be verified.

OPEN SOURCE ↗

[6] **ABDM Health Facility Registry**

National facility identifiers; PHC Pulse uses the HFR ID format as the facility key (synthetic IDs only).

OPEN SOURCE ↗

[7] **Google OR-Tools: minimum cost flow**

Open-source solver (SimpleMinCostFlow) used for redistribution under donor safety-stock limits.

OPEN SOURCE ↗

[8] **Gemini models on Vertex AI**

Gemini 2.5 Flash for stock extraction, register reading and transfer explanations.

OPEN SOURCE ↗